Does the Tobacco 21 Minimum Legal Sales Age Policy Improve Respiratory Health? Evidence from the United States

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Abstract

Tobacco use is a leading contributor to preventable respiratory morbidity, yet empirical evaluations of tobacco-control policies have largely focused on behavioral responses rather than health outcomes. We examine whether Tobacco-21 minimum legal sales age (MLSA) laws affect smoking and asthma prevalence among young adults aged 18–20. Using Behavioral Risk Factor Surveillance System data from 2011–2019, we exploit the staggered adoption of Tobacco-21 laws across U.S. states using the Callaway and Sant'Anna (2021) difference-in-differences estimator, which accommodates heterogeneous treatment timing. Smoking estimates are consistently negative, ranging from approximately 2 to 4 percentage points, although statistical significance varies across specifications. In contrast, we find no statistically detectable short-run effect on asthma prevalence. The asthma estimates are imprecise and do not rule out economically meaningful effects in either direction. These findings highlight the importance of distinguishing immediate behavioral responses from health outcomes that may require longer follow-up periods to detect.

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1 Introduction

Tobacco consumption is the leading cause of preventable death in the United States, contributing to more than 480,000 deaths per year (Center for Disease Control and Prevention, 2023). Of these deaths, about 29% are linked to respiratory causes, while 9% are due to exposure to secondhand smoke (Center for Disease Control and Prevention, 2020). Tobacco smoke accounts for a sizable share of asthma morbidity. Its direct use is responsible for about 20%–35% of asthma cases while secondhand exposure accounts for approximately 30%–50% of asthma symptoms in children (U.S. Department of Health and Human services, 2014). Tobacco use is also associated with an increased risk of heart disease, stroke, liver disease, and colon cancer (U.S. Department of Health and Human services, 2014). The healthcare costs are substantial as well, amounting to approximately $170 billion annually (Xu et al., 2015), with lifetime healthcare expenditures of around $86,000 for women and $183,000 for men who began smoking before the age of 24 (Sloan et al., 2006). Despite these well-documented adverse consequences of tobacco exposure, there is little empirical work on how tobacco regulation can improve respiratory health.

In this paper, we examine whether T21 MLSA laws reduce the prevalence of asthma among teenagers and young adults aged 18–20 in the United States. We focus on asthma for two reasons. First, it is the leading cause of chronic respiratory morbidity in this population, accounting for approximately 1.6 million emergency department visits and 140,000 hospitalizations annually (Center for Disease Control and Prevention, 2023). Second, asthma provides a useful setting for evaluating the extent to which changes in smoking behavior translate into observable health improvements. While reductions in tobacco use may lower exposure to respiratory irritants, asthma prevalence is a stock measure that reflects cumulative exposure and may respond slowly to short-term behavioral changes. Moreover, reductions in active smoking may be offset by continued exposure to secondhand smoke or changes in healthcare access and diagnosis, rendering the net effect of tobacco policy on asthma ambiguous.

Tobacco smoke contains numerous irritants that are harmful, and regular exposure to this smoke can potentially exacerbate asthma in humans (Tamimi et al., 2012; Strzelak et al., 2018). Experimental and clinical studies demonstrate that tobacco smoke introduces harmful chemicals into the lungs, impairing epithelial cells that line the airways and promoting excess mucus production (Strzelak et al., 2018; Tamimi et al., 2012). As a result, this impairs mucociliary clearance and reduces the lungs' ability to filter pathogens and irritants (Strzelak et al., 2018). Tobacco smoke also dysregulates local immune responses, provoking excessive inflammatory activity and further airway swelling and irritation (Strzelak et al., 2018). Inflammation of this type is similar to that observed in chronic respiratory diseases such as COPD (Tamimi et al., 2012). These biological changes narrow the airways and reduce airflow, thus making breathing more difficult.

The tobacco 21 MLSA law provides an interesting context for studying asthma among teenagers and young adults in the United States. Because most smokers in the United States initiate tobacco use before age 21, raising the minimum legal sales age directly targets the population at highest risk of initiation and dependence. According to the U.S. Surgeon General's report in 2012, approximately 96% of current smokers initiated smoking before the age of 21, and very few adults began smoking or transitioned to daily use after age 25 (U.S. Department of Health and Human services, 2018). Also, extensive research reveals that people who start smoking between the ages of 18 and 20 develop a stronger nicotine dependence and exhibit lower probabilities of cessation than those who start smoking at 21 years of age or later (Kwan et al., 2015; Abouk et al., 2024). Thus, the Tobacco 21 MLSA policy, which directly targets the age group at highest risk of smoking initiation, can potentially reduce access to tobacco use. If the policy reduces access to and use of tobacco among adolescents and young adults, it may lower exposure to cigarette smoke and improve respiratory health. However, if compliance is imperfect, access is maintained through informal channels, or individuals remain exposed to secondhand smoke, then reductions in smoking may not translate into measurable improvements in asthma prevalence, especially over short horizons.

A growing body of literature on the T21 MLSA laws finds that the policy reduced smoking participation among young people (see Bryan et al. 2020; Friedman and Wu 2020; Nuyts et al. 2020; Dai et al. 2021; Schiff et al. 2021; Colston et al. 2022; Grube et al. 2022; Liber et al. 2022; Wilhelm et al. 2022; Hansen et al. 2023; Patel et al. 2023; Abouk et al. 2024; Friedman and Pesko 2024; Cotti et al. 2024), however, whether these behavioral responses translate into short-term improvements in asthma prevalence remains an open empirical question. Although the Tobacco 21 minimum legal sales age (T21 MLSA) policy may lower the prevalence of asthma among teenagers and young adults by restricting their access to and exposure to cigarette smoke, the policy may not be effective for several reasons. First, a significant number of retailers do not comply with the MLSA regulations. Evidence from the literature suggests that about 62% of teenagers and young adults reported that it is "easy" or "somewhat easy" to purchase tobacco products in stores (Feng and Pesko, 2019; Mtenga and Pesko, 2024). Second, enforcement intensity and monitoring vary substantially across jurisdictions, potentially attenuating the policy's impact on asthma. Third, a sizable number of teenagers and young adults below the age of 21 rely on older peers for access to tobacco products or via third-party purchases (Abouk and Adams, 2017; Bryan et al., 2020). Moreover, some teenagers and young adults may be exposed regularly to secondhand smoke from older adults, which can potentially reduce the effects of the policy on asthma. Lastly, increase diagnosis as a result of increased access to health care could potentially attenuate the effects of the Tobacco 21 MLSA policy on asthma.

We hypothesize that increasing the minimum legal sales age reduces tobacco access among adolescents and young adults, leading to lower smoking initiation and intensity, and consequently improving respiratory health outcomes. To test this hypothesis, we use the 2011-2019 Behavioral Risk Factor Surveillance System (BRFSS) data. We explore the staggered adoption of the Tobacco 21 MLSA laws across states using the Callaway and Sant'Anna (2021) difference-in-difference estimation, which allows for treatment effect heterogeneity across cohorts and over time. We estimate the effects of the policy on both smoking behavior and asthma prevalence among individuals aged 18–20, the population most directly affected by the law. We also document the heterogeneity in the policy's effects on asthma across demographic groups.

Our results show that T21 MLSA laws lead to statistically significant reductions in smoking among young adults, consistent with prior evidence. However, we do not find statistically detectable effects on asthma prevalence in the short run. The estimated effects are small and imprecise, with confidence intervals that include both modest reductions and increases in asthma prevalence. These findings suggest that while T21 MLSA policies are effective at reducing smoking behavior, the resulting health benefits may not be immediately observable in chronic respiratory outcomes such as asthma.

We interpret these results through several mechanisms. First, asthma prevalence is a slow-moving outcome that reflects cumulative exposure, and short-run reductions in smoking may not immediately translate into changes in disease prevalence. Second, continued exposure to secondhand smoke may limit improvements in asthma morbidity even when individual smoking declines. Third, imperfect compliance with the policy and access through older peers may weaken its effectiveness in reducing overall exposure. Finally, changes in healthcare access and diagnosis associated with increase in insurance coverage may increase reported asthma prevalence and attenuate estimated policy effects.

Our paper contributes to the literature in three ways. First, we provide evidence that reductions in tobacco use induced by minimum legal sales age laws may not translate into measurable short-run improvements in asthma. While prior studies have documented robust effects of T21 MLSA policies on smoking behavior, we show that these behavioral changes do not necessarily translate into detectable changes in asthma prevalence over short horizons. Importantly, we interpret these null or imprecise estimates not as evidence of no health effect, but as reflecting limitations in measuring short-run health responses using stock outcomes.

Second, we contribute a framework for understanding the relationship between behavioral and health responses to public health policies. We highlight how factors such as secondhand smoke exposure, imperfect compliance, and the biological dynamics of chronic disease can weaken the link between reductions in risky behavior and observable health outcomes. This framework helps reconcile why policies that meaningfully change behavior may yield limited or delayed effects on commonly used health measures.

Third, our findings speak to the broader evaluation of public health interventions. We show that reliance on short-run changes in chronic health outcomes may understate the benefits of tobacco control policies. This underscores the importance of aligning outcome measures with the time horizon over which policy effects are expected to materialize and suggests that more responsive outcomes such as symptoms, or healthcare utilization may be better suited for detecting short-run health impacts. Thus, our findings suggest that the absence of short-run changes in chronic health outcomes does not imply that a policy is ineffective, but may instead reflect the timing, measurement, and pathways through which health improvements occur.

The remainder of the paper is organized as follows. Section 2 reviews the Tobacco 21 MLSA policy in the United States and summarizes the link between smoking and asthma. Section 3 describes the data. Section 4 discusses the empirical strategy, section 5 presents and analyzes the results and section 6 concludes the paper.

2 Background to Tobacco 21 Minimum Legal Sales Age Policy in the United States

In an effort to reduce the growing number of teenage smokers and its health effects, several anti-smoking policies such as tax hikes, warning labels and minimum legal sales age (MLSA) were introduced in the 1990s and 2000s as a measure to restrict teenagers from accessing tobacco products (Bryan et al., 2020). Although the tobacco MLSA laws in the United States were first introduced in late 19th century, the history of the current Tobacco 21 law dates back to 1996 when a non-profit organization—Preventing Tobacco Addiction Foundation, first launched a campaign to raise the tobacco minimum legal sales age to 21. The campaign, which excluded nicotine replacement therapy items like gums and patches, however, covered all nicotine-based products at the time. The initiative was targeted at reducing tobacco addiction among young people by limiting their access to these products during their formative years (Abouk et al., 2024).This campaign marks the beginning of organized efforts to promote T21 legislation in the United States.

Nine years after the campaign, Needham—a municipal in the state of Massachusetts (M.A.), became the first municipal in the United States to raise the minimum tobacco sales age from 18 to 21. The local ordinance, which was known as "Tobacco 21," aims to reduce access to tobacco products for adolescents and young adults in Needham, MA. As of May 2016, the Tobacco 21 policy has been enacted in over 130 cities and towns across nine states. These include major metropolitan areas like New York City and Chicago. At the state-wide level, seventeen states implemented the tobacco policy. California and Hawaii were the first to raise the MLSA of tobacco from 18 to 21 in 2016, followed by New Jersey and Washington DC in 2017. Three states (OR, ME, and M.A.) later joined in 2018, and ten other states (V.A., IL, DE, AR, MD, VT, TX, CT, NY, and O.H.) followed in 2019. However, certain states' tobacco MLSA laws include a grandfather provision which allows under aged between 18 and 21 to continue to buy tobacco products (Abouk et al., 2024). For example, in Texas, the provision permits individuals aged 18 to 20 at the time of the enactment of Tobacco 21 (T21 hereafter) to continue to purchase tobacco, while in Maryland, the laws permit tobacco sales to underaged individuals actively serving in the military (Abouk et al., 2024).

More recently, the United States government amended the Federal Food, Drug, and Cosmetic Act in 2019 to raise the tobacco MLSA from 18 to 21. The T21 law, as it is popularly called, prohibits the sale of tobacco products such as cigarettes, smokeless tobacco, hookah tobacco, cigars, pipe tobacco, and electronic nicotine delivery systems (including e-cigarettes and e-liquids) to individuals under the age of 21. Despite the federal amendment of the tobacco MLSA, its full enforcement hinges on the publication of a Final Rule, which has not been issued as of November 2022 (Abouk et al., 2024). This ambiguity regarding federal enforcement has prompted many localities and states to independently enact their T21 regulations following its enactment at the federal level.

Shortly after the federal law was passed in December 2019, thirteen states (W.A., UT, PA, SD, WY, IN, MN, OK, IA, MS, CO, NH, and N.E.) adopted the T21 MLSA policy in 2020 and ten other states (N.M., KY, TN, GA, ND, FL, AL, NV, LA, and R.I.) followed in 2021. Two states (I.D. and MI) joined in 2022, and lastly, Kansas State joined in July 2023 (see https://tobacco21.org). As of December 2024, eight states (S.C., NC, AZ, MT, WI, AK, and MO) are yet to enact state law raising the tobacco MLSA from age 18 to 21 (see Figure 1 and Appendix Table A.1 for map and effective date of states' adoption of T21 MLSA policy). In this paper, we provide evidence regarding the potential effects of T21 MLSA policy on the prevalence of asthma in states that are yet to adopt the T21 MLSA policy beyond 2019.

3 Data Description

We employ the 2011-2019 Behavioral Risk Factor Surveillance System (BRFSS) dataset to provide a causal estimate. The BRFSS is a repeated cross-section state representative annual telephone survey administered to individuals who are 18 years and above. The survey is jointly conducted by the Center for Disease Control and Prevention (CDC) and state authorities, covering a wide range of factors, including tobacco use and respiratory health outcomes. It is administered every month in all 50 states and the District of Columbia through random number dialing. The BRFSS survey is designed to represent the non-institutionalized adult population of the United States.

The advantage of BRFSS in this study is that it covers a wide range of outcomes of interest, including asthma, smoking patterns, and other self-reported health indicators. Additionally, it contains state-specific identifiers and demographic information which makes it easy to isolate respondents by treatment and control states. With a large sample size between adolescence and older adults of nearly 500,000 individuals surveyed each year, the BRFSS significantly contains a significant sample size of young adults from 18 to 20 years old that are affected by the Tobacco 21 MLSA law. In comparison, the Youth Risk Behavioral Risk Surveillance System (YRBSS) and the National Youth Tobacco Survey (NYTS) have an annual sample size of nearly 50,000 and 30,000 individuals, respectively, which is less than one-tenth of the BRFSS sample size. In addition, the YRBSS and NYTS have fewer samples of respondents aged 18 and above, which may be inadequate to provide a plausible estimate of the effects of Tobacco 21 MLSA policy on the prevalence of asthma. To estimate the effect of the T21 MLSA policy on the prevalence of asthma, we construct a binary indicator for the outcome using the BRFSS metric. Individuals who respond "Yes" to have asthma at the time of the interview are assign 1, whereas those who respond "No" are assign 0. Those who respond "don't know/not sure" or" refused" are code as missing. Additionally, we construct a binary variable for current smokers using the BRFSS metric. Individuals who answered "Yes" to currently smoke tobacco are coded as 1, while those who answered "No" to smoking are coded as 0. Responses such as "Don't know/Not sure" or "Refused" are treated as missing data.

In this paper, we utilize the BRFSS data from 2011 to 2019. We exclude earlier years due to a methodological redesign of the survey and later years to avoid confounding effects associated with the COVID-19 pandemic, which significantly affected respiratory health outcomes. This time helps mitigate concerns about confounding factors, although such concerns cannot be fully ruled out. We restrict the BRFSS sample to the demography targeted by the T21 MLSA policy: adults under 21. The condition for inclusion in the main analysis is that a respondent must be aged 18 to 20. We defined treatment states as those states that have adopted the T21 MLSA by 2019. States without the Tobacco 21 MLSA policy are classified under the control group. We exclude six states with local Tobacco 21 MLSA policy prior to state-wide implementation. These states include CA, IL, MA, MO, NY, and OH. Table 1 presents the demographic characteristics of the sample used in the main analysis.

4 Empirical Strategy

4.1 Callaway and Sant'Anna (2021) Difference-in-Difference

This paper seeks to identify the causal impact of the Tobacco 21 Minimum Legal Sales Age (T21 MLSA) on the prevalence of asthma using data from the Behavioral Risk Factor Surveillance System (BRFSS) spanning from 2011 to 2019. However, our research faces the issue of the gradual roll-out of the T21 MLSA policy across states in the United States as shown in Figure 1 and Table A.1. In such a setting with treatment effect heterogeneity, the existing econometric literature argued that the estimates from the canonical difference-in-differences (static two-way fixed effects) and event-study (dynamic two-way fixed effects) methods may be biased due to negative weighting and treatment effect heterogeneity (de Chaisemartin and d'Haultfoeuille, 2020; Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021; Sun and Abraham, 2021). We therefore implement the Callaway and Sant'Anna (2021) estimator, which allows for fully flexible treatment effects by cohort and event time. Specifically, we use the the doubly robust (DR) estimator which combines regression outcome and inverse probability weighting (IPW).

The Callaway and Sant'Anna (2021) approach is based on two key assumptions. First, it assumed that in the absence of treatment, outcomes for all cohorts follow parallel trends during the post-treatment period. In addition, it is more flexible and robust to model misspecification and does not require aggregating group specific estimates, unlike the imputation estimator, which is also robust to heterogeneous treatment but requires aggregation across group specific estimates and correct model specification (Borusyak et al., 2021). The second identification assumption is that the treatment has no anticipation effect on the outcome variable. Thus, in the absence of the anticipation effect, the estimator assumed that the treatment had no causal effect on outcomes before implementation. The Callaway and Sant'Anna (2021) DiD estimator also allows for separate identification of average treatment effects on the treated for each treatment cohort g and calendar year t, denoted as $\mathrm{ATT}(g, t)$. The $\mathrm{ATT}(g, t)$ is derived by comparing the average change in outcome between periods $g - 1$ and t for cohort g with the average change over the same period for the control group. Using never-treated states as the control group in one case, along with both never-treated and not-yet-treated states in another case, we estimate the effect of the T21 MLSA policy on the prevalence of asthma using equation 1 below:

$$ATT(g, t) = \mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1, X_{ist}] \tag{1}$$

Where $\mathrm{ATT}(g, t)$ is the difference between the potential outcomes before and after the units in that group are treated. The variable X_{st} is the vector of socio-economic demographic characteristics of the respondents which includes age, sex, marital status, educational level, race, and employment status. We also include time varying state controls such as cigarette taxes and indoor smoking policy which could potentially affect the prevalence of asthma. After the estimation, we aggregated the individual estimates $\mathrm{ATT}(g, t)$ values to derive an economically relevant parameter through their weighted average as displayed in Table 2. Following Callaway and Sant'Anna (2021), we present the standard errors using a multiplicative bootstrap procedure with clustering at the state level. We also incorporate BRFSS sampling weight in our estimation to preserve representativeness. We reported the group-specific estimates and partially aggregated estimates weighted for all the cohorts. First, in the absence of treatment, outcomes for treated and comparison states would have followed parallel trends.

Our identification strategy exploits the staggered adoption of Tobacco-21 MLSA laws across U.S. states and relies on a difference-in-difference framework. The key identifying assumption is that, in the absence of policy adoption, asthma prevalence and smoking behavior for treated and comparison states would have followed parallel trends. To evaluate this assumption, we estimate event-study specifications and examine pre-treatment trends. We also employ recently developed DiD estimators that are robust to heterogeneous treatment timing and effect, following Callaway and Sant'Anna (2021) and Sun and Abraham (2021). These approaches reduce the risk of bias associated with traditional two-way fixed effects estimators when treatment effects vary across cohorts or over time. We also test for parallel trends using the event study estimates in equation 2 for n years relative to the T21 MLSA policy adoption for $n = \{-5, \ldots, -2, 0, \ldots, 1, 2, 3\}$. The key identifying assumption is that, in the absence of policy adoption, trends in asthma prevalence and smoking behavior would have evolved similarly across treatment and control states. To evaluate this assumption, we estimate the following event-study equation:

$$ATT_n^{ES} = \sum_g w(g, n)\mathrm{ATT}(g, g+n). \tag{2}$$

Where $w(g, n)$ is the weight determined based on the relative size of the group g among the groups that have participated in the treatment for n periods, which enables an aggregation that facilitates the examination of dynamic policy effects. While the ATT_n^{ES} is the average treatment on treated given the event study (ES) at n years. In our event study, we find no statistically significant differential pre-trends between treated and control states in the five years preceding adoption.

To validate our estimate, we perform a falsification test on individuals aged 24–28 years who are not affected by the implementation of the T21 MLSA policy. If the identifying assumptions hold, we should not observe any significant change in the prevalence of asthma for this group before and after the policy change. Finding no effect would support the validity of our main empirical strategy. However, a significant effect change for this group would suggest the presence of simultaneous changes in other policies, which could bias our main estimates. Additionally, we conduct a placebo test with outcome variables that should be theoretically unaffected by the implementation of the T21 MLSA law—respondents' unemployment and household income. Also, finding a strong effect of the T21 MLSA law on these outcomes would also indicate that other policy changes occurring either in the treatment or control group could bias the main estimate. Additionally, We perform a heterogeneous test by estimating the regression model separately for different subgroups of respondents by sex, race, education, employment, and age.

There is also the concern that policy implementation and enforcement may vary within states. For example, if compliance differs across local jurisdictions, it is likely to attenuate estimated effects toward zero. However, we rule out this concern by excluding states with prior local policies. In addition, we perform sensitivity analysis, including states with local Tobacco 21 MLSA laws prior to their respective statewide implementation. We estimate the regression model using the two-way fixed effects and Callaway and Sant'Anna (2021) difference-in-difference method to show whether our main estimates are sensitive to excluding the six states with local tobacco 21 MLSA policies before the statewide policy change.

Lastly, there is the concern that other state-level tobacco control policies implemented during the same period could confound our estimates of the Tobacco 21 MLSA effects. Between 2011 and 2019, many states enacted additional tobacco regulations, such as increases in cigarette excise taxes, smoke-free air laws, and vaping restriction laws, that may independently affect smoking behavior and respiratory health. If these policies are correlated with the timing of Tobacco-21 adoption, failing to account for them could bias our results. To address this, our preferred specifications also include controls for key time-varying state tobacco policies, such as cigarette taxes and indoor smoking bans. Incorporating these controls does not change our estimates, suggesting that our findings are not driven by other concurrent tobacco policies. That said, we cannot rule out the possibility that unobserved or imperfectly measured policies may still confound our results, and future research using more detailed policy data could help address this limitation.

5 Results and Discussion

5.1 Main Results

We begin our analysis with the descriptive statistics of the outcome variable and the demographic characteristics of the sample used in this paper as presented in Table 1. We examined the sample mean of the outcome variable (asthma) from 2011 to 2019 and observed no substantial difference between the treated and control states. We find no effects of Tobacco 21 MLSA policy on the prevalence of asthma among adolescents and young adults aged 18 to 20 in the treated states compared to their counterpart in the control states. However, the sample means for individual demographic characteristics are largely comparable between the treated and control states, with the exception of a modest difference in the proportion of white individuals. We also observed a slight difference in the indoor smoke-free laws between the two groups.

Table 2 reports the main estimates of the effect of the Tobacco 21 MLSA law on the probability that individuals aged 18 to 20 report current asthma. Columns 1 and 2 present results from the two-way fixed effects (TWFE) specification, while columns 3–6 report estimates based on the Callaway and Sant'Anna (2021) approach. Columns 3 and 4 use never-treated states as the comparison group and impose no anticipation effects, whereas columns 5 and 6 use not-yet-treated states as the comparison group and allow for a one-year anticipation window. All specifications include state and year fixed effects. Columns 2, 4, and 6 additionally control for individual characteristics and other time-varying state tobacco policies. Our preferred specification, reported in column 4, uses the Callaway and Sant'Anna (2021) estimator with never-treated states as the comparison group and includes the full set of controls.

Using both the two-way fixed effects (TWFE) specification (Columns 1–2) and the Callaway and Sant'Anna (2021) estimator (Columns 3–6), we find no statistically detectable short-run effects of Tobacco-21 on asthma prevalence among 18–20-year-olds. Across specifications, the estimates are small and imprecisely estimated, with confidence intervals that include both modest reductions and increases in asthma prevalence. Given the standard errors, the minimum detectable effect is approximately 14 percentage points at conventional levels of statistical power, implying that the estimates do not rule out economically meaningful effects but may lack power to detect small-to-moderate changes. These results are robust to the inclusion of individual covariates and other state-level tobacco control policies, and should be interpreted with caution. Accordingly, the null estimates should be interpreted as inconclusive rather than evidence of no effect.

Table 3 presents the results for the effects of the Tobacco 21 MLSA policy on the probability of teenagers and young adults aged 18 to 20 who currently smoke. Using our preferred Callaway and Sant'Anna (2021) DiD estimation in columns 3, we find that the Tobacco 21 MLSA policy reduces the probability of teenagers and young adults who currently smoke. In particular, we estimate that the Tobacco MLSA policy significantly decreases teenagers' and young adults' current smoking by 4 percentage points (67%; $p < 0.05$). Almost all the TWFE and Callaway and Sant'Anna (2021) DiD estimates are statistically significant with similar signs and magnitude. Although Tobacco-21 MLSA policy significantly reduced youth smoking, we find no statistically detectable short-run effects on asthma prevalence. This finding suggests that reductions in smoking do not immediately translate into measurable respiratory health gains, consistent with medical evidence that lung recovery and changes in chronic respiratory conditions often occur with substantial delay. Thus, emphasizing the importance of distinguishing between behavioral and health responses when evaluating tobacco control policies.

Figure 2 and 3 show the event study for asthma and smoking outcomes respectively. Both the treatment and control states followed parallel trends in the pre-2016 period, which lends credence to the causal interpretation of the average treatment effects estimated in Table 2 and Table 3. Although it may seem reasonable to think that teenagers and young adults from aged 18 to 20 may anticipate the increase in the Tobacco minimum legal sales age from 18 to 21 and may try to buy and use more tobacco products in advance, which could potentially worsen the prevalence rate of asthma before the policy reform, the event study estimates presented in Figure 2 show no evidence of such anticipatory behavior before the 2016 implementation of the Tobacco 21 policy. We fail to reject the null hypothesis that all pre-treatment coefficients are jointly equal to zero ($p > 0.10$). Furthermore, estimates from our event study show no decrease in the prevalence of asthma in treated states after the policy change. The estimates are not statistically significant at 10% level, suggesting that the policy change have little or no effect on the prevalence of asthma among teenagers and young adults, supporting our causal interpretation in Table 2. On the other hand, our event study estimates for smoking outcome in Figure 3 show a significant decrease in the probability of smoking following the implementation of the Tobacco 21 MLSA policy in the treatment states. As shown in Figure 3 our estimates are statistically significant at 10% level which lend credence to our causal interpretation in Table 3.

Tables B.1 and B.2 in the Appendix report group–time average treatment effect estimates of the Tobacco 21 minimum legal sales age (MLSA) policy on asthma prevalence using the never-treated and not-yet-treated comparison groups, respectively. Across both control groups, the estimated effects are small and statistically indistinguishable from zero. In the never-treated specification, none of the eight group–time estimates indicates a detectable effect of the policy on asthma prevalence among individuals aged 18–20. The average treatment effect across treated states is approximately five percentage points, but this estimate is economically modest and statistically imprecise. Our results using the not-yet-treated comparison group are similar: group–time estimates range from 0.00 to 0.21 percentage points and are not statistically distinguishable from zero. Taken together, these findings indicate that the Tobacco 21 MLSA policy has no detectable short-term effect on asthma prevalence in this population.

Similarly, Table B.3 and B.4 in the appendix present the detail results of the effects of the Tobacco 21 MLSA on current smoking for the never treated control and not yet treated control respectively. The group-time average treatment effects for the never treated comparison group also provide support for the view that the Tobacco 21 MLSA law has significant effects on current smoking rate among teenagers and young adults aged 18 to 20. Two (2) out of 8 group-time average treatment effects, shows a clear negative and statistically significant effects of Tobacco 21 MLSA law on smoking. Three are marginally insignificant and positive, while the other three are negative and statistically insignificant. The group time average treatment effect ranges from 0.0% to 10.3% reduction in the probability of smoking among teenagers and young adults. While the average treatment effect of raising the Tobacco minimum legal sales age from 18 to 21 across all the treated states is 3.6%. Similarly, in the case of the not yet treated comparison group, we find Tobacco 21 MLSA policy effects in 2 out of 8 group time average treatment effects to be statistically significant in decreasing the probability of smoking among teenagers and young adults aged 18 to 20. Three are negative and marginally insignificant. The other three are positive and statistically insignificant. In the case of the not yet treated, the group time effect ranges from 0.4% to 9.4% decrease in the probability of smoking among teenagers and young adults aged 18 to 20. The average effect of Tobacco 21 MLSA law on asthma across all the treated states is 3.7%.

Although our estimates show that the Tobacco 21 MLSA policy reduces the probability of teenagers and young adults aged 18 to 20 who currently smoke, our main results show that this effect does not translate to a significant reduction in the prevalence of asthma among this age group. The lack of significant short-term effects of the Tobacco 21 MLSA policy on asthma may be due to several factors. First, in an effort to maximize profits, many retailers do not comply with the MLSA regulations. The literature reports that about 62% of teenagers and young adults reported that it is "easy" or "somewhat easy" to purchase tobacco products in stores (Feng and Pesko, 2019; Mtenga and Pesko, 2024), suggesting that regulatory constraints have not been effective in preventing point-of-sale access to tobacco. As a result, the policy's capacity to meaningfully curb youth access and exposure to tobacco products remains limited.

Secondly, selling tobacco to minors may appear advantageous to retailers, as fines are typically smaller than the potential revenue from such sales. According to Abouk et al. (2024), retailers who violate the Tobacco MLSA law are subject to a civil penalty of not more than $500, and only in rare cases are their licenses withdrawn. Consequently, some retailers may perceive non-compliance as a low-risk, high-reward decision. This dynamic can weaken the policy's deterrent effect and reduce its overall effectiveness. Thirdly, sting operations intended to prevent underage sales are often ineffective, as underage decoys are usually prohibited from employing common strategies used by actual underage buyers, such as presenting fake identification or misrepresenting their age (Mtenga and Pesko, 2024). This challenge creates a discrepancy between experimental compliance checks and real-world purchasing behavior. Consequently, enforcement metrics may overstate true compliance and underestimate youth access to tobacco products. Also, a sizable number of teenagers and young adults below the age of 21 rely on older peers for access to tobacco products or via third-party purchases (Abouk and Adams, 2017; Bryan et al., 2020). These alternative procurement channels allow minors to circumvent retail restrictions entirely. Hence, even strict point-of-sale enforcement may have limited influence on actual youth consumption patterns. Thus, it may render the Tobacco 21 MLSA law ineffective.

Third, the lack of effects may be due to the implementation of the Affordable Care Act (ACA) Medicaid expansion across states between 2014 and 2019, which significantly increased access to healthcare and diagnosis. This expansion may have increased the likelihood of asthma detection and reporting, thereby attenuating the measured effects of the Tobacco 21 MLSA policy. If diagnosis rates rise independently of underlying health improvements, estimated treatment effects may be biased toward zero.

Also, the insignificant effects of the Tobacco MLSA policy on asthma could be because some teenagers and young adults may be exposed regularly to secondhand smoke from older adults, which could be detrimental to their respiratory health. Such exposure may persist irrespective of legal purchasing restrictions.

Consequently, improvements in youth respiratory health may be muted, even if direct tobacco consumption declines. This mechanism complicates causal attribution and may attenuate the observed effects of the Tobacco 21 MLSA policy on asthma prevalence.

Lastly, asthma is a chronic respiratory disease whose prevalence and severity may respond slowly to changes in smoking behavior. While reductions in tobacco use can immediately decrease exposure to respiratory irritants, measurable improvements in asthma prevalence may require longer restriction periods for teenagers and young adults before physiological benefits are realized. Consequently, short-term policy evaluations may underestimate the full health impact of tobacco access restrictions suggesting that future research using longer post-implementation windows may yield more precise estimates for asthma outcome.

5.2 Heterogeneity in Tobacco 21 Minimum Legal Sales Policy

Table 4 and Figure B.2 present the results from our heterogeneous analysis. Overall, we find no evidence that the Tobacco 21 MLSA policy led to a statistically detectable reduction in asthma prevalence across most subgroups. However, point estimates for Black respondents and for individuals aged 19 suggest larger reductions in asthma prevalence relative to other groups, although these estimates are statistically imprecise. For Black teenagers and young adults, the estimates indicate a 10 percentage point reduction in asthma prevalence, corresponding to approximately a 77 percent decrease relative to the pre-reform mean. Similarly, for individuals aged 19, we estimate a 6 percentage point reduction in asthma prevalence, equivalent to roughly a 50 percent decline. Although point estimates suggest larger reductions for Black respondents and 19-year-olds, these estimates are imprecise and should be interpreted cautiously. The lack of precision may reflect limited statistical power, measurement error in self-reported asthma, or delays in the relationship between changes in smoking behavior and observable health outcomes.

The subgroup estimates presented here are exploratory and should be interpreted with caution, as they were not pre-specified and are not statistically distinguishable from zero. Observed heterogeneity may reflect underlying differences in asthma burden, tobacco exposure patterns, or behavioral responses across demographic groups. For example, non-Hispanic Black individuals bear a disproportionately high asthma burden and may face higher exposure to secondhand smoke, potentially making asthma outcomes more sensitive to reductions in smoking behavior; similar sociodemographic differences in smoking initiation and peer influence have been documented in the literature (Forno and Celedón, 2012). Nevertheless, given the imprecision of these subgroup estimates, definitive conclusions about differential effects cannot be drawn and warrant further investigation.

5.3 Mechanism

A direct channel through which the Tobacco 21 MLSA law can reduce the prevalence of asthma is by limiting teenagers' and young adults' access to tobacco products. To assess the policy's effectiveness, we use BRFSS data from 2011 to 2019 and examine how raising the minimum sales age affects smoking rates. Using staggered difference-in-difference (DiD) and TWFE methods, we compare trends in smoking among teens and young adults in states that implemented the policy versus those that did not. Consistent with prior work, including Abouk et al. (2024), we document a statistically significant decline in smoking among individuals aged 18–20 in treatment states following implementation (see Appendix Table 3). Despite the reduction in smoking behavior, we do not observe a statistically detectable effect on asthma prevalence in the short run. Although raising the minimum sales age limit legal access to tobacco, its effectiveness depends on strong enforcement. The literature reports that about 62% of teenagers and young adults reported that it is "easy" or "somewhat easy" to purchase tobacco products in stores (Feng and Pesko, 2019; Mtenga and Pesko, 2024), suggesting that regulatory constraints have not been effective in preventing point-of-sale access to tobacco. In such cases, Tobacco 21 might fall short in preventing early tobacco use and reducing short-term respiratory health risks like asthma.

Even when the Tobacco-21 MLSA policy successfully reduces direct smoking among adolescents and young adults, many individuals may still be exposed to secondhand smoke in homes, workplaces, or public spaces. For example, exposure to secondhand smoke is responsible for about 30%–50% of asthma symptoms in children (U.S. Department of Health and Human services, 2014). Such exposure is known to exacerbate respiratory symptoms and trigger asthma attacks, particularly among vulnerable populations. Consequently, reductions in personal tobacco use may not translate into proportional declines in overall smoke exposure, limiting the short-term health benefits of the policy. This channel is particularly relevant for young adults who live with older smokers or reside in communities with higher smoking prevalence.

Another important mechanism relates to the biological timing of respiratory health responses. Asthma is a chronic inflammatory condition shaped by cumulative exposure to environmental irritants. While reductions in smoking can immediately lower exposure to harmful substances, measurable improvements in asthma prevalence may require longer periods of reduced exposure. Structural changes in the airways, inflammatory patterns, and immune responses do not necessarily reverse quickly following behavioral change. As a result, short-term policy evaluations may underestimate the full health impact of tobacco control measures, even when they successfully reduce smoking.

Taken together, these mechanisms suggest that the absence of a statistically detectable short-run effect of Tobacco-21 MLSA policy on asthma prevalence does not imply that the policy is ineffective for respiratory health. Rather, it highlights the importance of considering compliance, environmental exposure, increased diagnosis, and biological timing when evaluating the health consequences of tobacco regulations. More broadly, our results suggest that short-run evaluations of tobacco policies based solely on health outcomes may understate their effectiveness, particularly when outcomes evolve over longer horizons. Our findings are therefore consistent with a scenario in which Tobacco-21 MLSA policy reduces smoking behavior, but measurable health benefits may emerge only over longer horizons. Although our study period spanned 2011 to 2019, the first statewide implementation of the policy did not happen until 2016, leaving only four years of post-policy data for the earliest adopting states. Thus, we are unable to estimate the long-term impact of the policy on asthma because of the COVID-19 pandemic, which significantly affected respiratory health outcomes in 2020. This limitation made it difficult to extend the study period without introducing major external disruptions to the data. Therefore, we concluded our analysis in 2019 to avoid confounding effects from the pandemic. Future studies could account for COVID-19 effects and extend the scope beyond 2019.

5.4 Robustness Checks

Table B.5 and Table B.7 in the appendix present the results for the placebo test and falsification test, respectively. We conducted a placebo test for outcome variables such as unemployment and household income, which should be theoretically uncorrelated with the Tobacco 21 MLSA law. As expected, almost all the average treatment effects for both placebo outcomes are positive and statistically insignificant (see Appendix Table B.5). These findings provide reassurance that unrelated macroeconomic trends do not influence the sign and magnitude of our main estimate. Overall, the placebo results support the validity of our identification strategy. Similarly, we present the falsification tests for young adults aged 24–28 years, a group unaffected by the Tobacco 21 MLSA policy (see Appendix Table B.6). For this cohort, we find that the average treatment effect is close to zero and statistically insignificant, which increases our confidence that no other differential policy changes occurred in the treatment and control states concurrently with the implementation of the Tobacco 21 MLSA policy.

In addition, there is the concern that our state-level analysis may mask meaningful within-state variation in the enforcement of Tobacco-21 laws and in actual exposure to tobacco smoke. For example, if compliance varies across localities, our estimates are likely to reflect an average of high- and low-enforcement areas, which could attenuate the policy's effects toward zero and make it more difficult to detect true impacts on respiratory health. We address this concern by excluding states where there were local tobacco 21 MLSA law in place before statewide adoption in our main analysis. The restriction helps mitigate bias or concern that may arise from early local implementation within states and provides a cleaner comparison of pre- and post-policy outcomes. In addition, we conducted a sensitivity analysis as shown in Table B.7 of the appendix to test whether our estimates are sensitive to the inclusion of state (such as CA, IL, MA, MO, NY, and OH) with local Tobacco 21 MLSA laws prior to statewide implementation. The results remain consistent with our main estimates in Table 2, suggesting that the results are not driven by within-state variation arising from earlier local policies.

Although we find no evidence that the Tobacco MLSA policy reduces asthma prevalence, our analysis is subject to several limitations. First, we rely on BRFSS data, which are self-reported and contain substantial missing information or questions were not asked on immediate asthma-related outcomes, such as symptoms, diagnoses, episodes, emergency visits, and physician visits. Consequently, we focus on current asthma prevalence, which captures short-run outcomes but may not reflect more immediate health responses. Second, we are unable to directly observe individual exposure to secondhand smoke at high frequency, which limits our ability to disentangle behavioral from environmental channels. Third, variation in enforcement and compliance with the Tobacco-21 MLSA policy within states is not directly observed, making it difficult to account for differences in treatment intensity. Finally, despite controlling for a rich set of covariates and fixed effects, unobserved time-varying factors correlated with both policy adoption and respiratory health may bias our estimates. Accordingly, our findings should be interpreted with caution.

6 Concluding Remarks

This study examines the short-term respiratory health implications of staggered state adoption of Tobacco-21 MLSA laws in the United States. Using nationally representative BRFSS data and modern difference-in-difference estimators, we find consistent evidence that the policy significantly reduces smoking behavior among young adults aged 18–20. While Tobacco-21 MLSA policies are effective in reducing smoking among young adults, our results indicate that these behavioral changes do not translate into detectable short-run improvements in asthma prevalence. This suggests an important disconnect between behavioral and health responses, particularly for chronic conditions that evolve over time.

The evidence from our study offers policy-relevant guidance for states and countries seeking to address rising respiratory health concerns associated with tobacco exposure among teenagers and young adults. While policymakers may continue to employ Tobacco 21 MLSA policies as a tool for restricting youth access to tobacco products, complementary strategies are likely needed to enhance their effectiveness. In particular, stricter enforcement mechanisms and higher penalties for retailer violations could strengthen compliance and reduce underage sales. Beyond MLSA regulations, jurisdictions may also consider expanding indoor smoke-free policies for adults and creating designated smoking areas to limit secondhand smoke exposure among young people. Our findings contribute to the broader literature by demonstrating that policy evaluations based on short-run health outcomes may fail to capture the full benefits of tobacco control interventions.

Table 1: Summary Statistics (Source: Behavioral Risk Factor Surveillance System (BRFSS), 2011 to 2019)

	Treatment (1)	Control (2)	T-test Difference (3)
Outcome			
Asthma (0/1)	0.13	0.11	0.02
	(0.35)	(0.32)	(0.03)
Current Smoker (0/1)	0.01	0.05	-0.04
	(0.09)	(0.22)	(0.02)
Demographic Characteristics			
Income ($)	66,296.30	55,919.53	10,376.77
	(35,185.88)	(34,381.49)	(2,979.54)
Unmarried (0/1)	0.99	0.96	0.03
	(0.09)	(0.19)	(0.02)
Age (yrs)	18.84	18.90	-0.06
	(0.81)	(0.82)	(0.07)
Male (0/1)	0.47	0.54	-0.07
	(0.50)	(0.50)	(0.04)
Female (0/1)	0.53	0.46	0.07
	(0.50)	(0.50)	(0.043)
Less than High School (0/1)	0.10	0.15	-0.05
	(0.31)	(0.36)	(0.03)
High School (0/1)	0.47	0.50	-0.02
	(0.50)	(0.50)	(0.04)
Some College (0/1)	0.42	0.34	0.08
	(0.50)	(0.47)	(0.04)
College and advance (0/1)	0.00	0.01	-0.01
	(0.00)	(0.11)	(0.01)
White (0/1)	0.57	0.67	-0.13
	(0.50)	(0.47)	(0.03)
Black (0/1)	0.12	0.10	0.02
	(0.32)	(0.30)	(0.02)
Hispanic (0/1)	0.13	0.12	0.09
	(0.33)	(0.32)	(0.02)
Other race (0/1)	0.17	0.11	0.04
	(0.38)	(0.43)	(0.02)
Self-employed (0/1)	0.03	0.03	-0.01
	(0.17)	(0.18)	(0.02)
Working (0/1)	0.35	0.36	-0.01
	(0.48)	(0.48)	(0.04)
State Cigarette tax (0–1)	0.06	0.01	0.04
	(0.16)	(0.09)	(0.01)
State Indoor Smoke law (0/1)	0.88	0.74	0.14
	(0.27)	(0.38)	(0.03)

Note: Table displays the summary statistics of the outcomes and demographic characteristics for the treated and untreated states.Young adults are observed at ages 18-20 years between 2011 and 2019. Each cell reports weighted mean with standard deviation in parenthesis

Table 2: Main Results for the Effect of T21 MLSA Policy on Asthma (Source: BRFSS, 2011 to 2019)

			Never Treated		Not yet Treated	
	TWFE	TWFE	C & S	C & S	C & S	C & S
	(1)	(2)	(3)	(4)	(5)	(6)
T21Policy	0.02	0.02	0.05	0.01	0.05	0.02
	(0.03)	(0.03)	(0.05)	(0.08)	(0.05)	(0.07)
Pre-2016 Mean	0.11	0.11	0.11	0.11	0.11	0.11
Sample Size	10,220	10,220	10,220	10,220	10,220	10,220
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes

Note: The first-two columns presents the simple average of the event-time effects obtained using the Two-way Fixed effect estimator. While Column 3-6 present the average treatment effect for the Callaway and Sant'Anna (2021) estimator for not yet treated and never treated controls. Column 3 and 4 display the estimates for the never treated controls, while column 5 and 6 presents that of the not yet treated controls. Young adults are observed at ages 18-20 years between 2011 and 2019 asthma outcome. Unless otherwise stated all regressions control for state, and year fixed effects. In addition, we control for individual characteristics and other state tobacco policies ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table 3: The Effect of T21 MLSA Policy on current Smoking (Source: BRFSS, 2011 to 2019)

			Never Treated		Not yet Treated	
	TWFE	TWFE	C & S	C & S	C & S	C & S
	(1)	(2)	(3)	(4)	(5)	(6)
T21Policy	-0.04**	-0.03**	-0.04**	-0.02	-0.04**	-0.02
	(0.01)	(0.01)	(0.02)	(0.02)	(0.02)	(0.02)
Pre-2016 Mean	0.06	0.06	0.06	0.06	0.06	0.06
Sample Size	9,384	9,384	9,384	9,276	9,384	9,384
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes

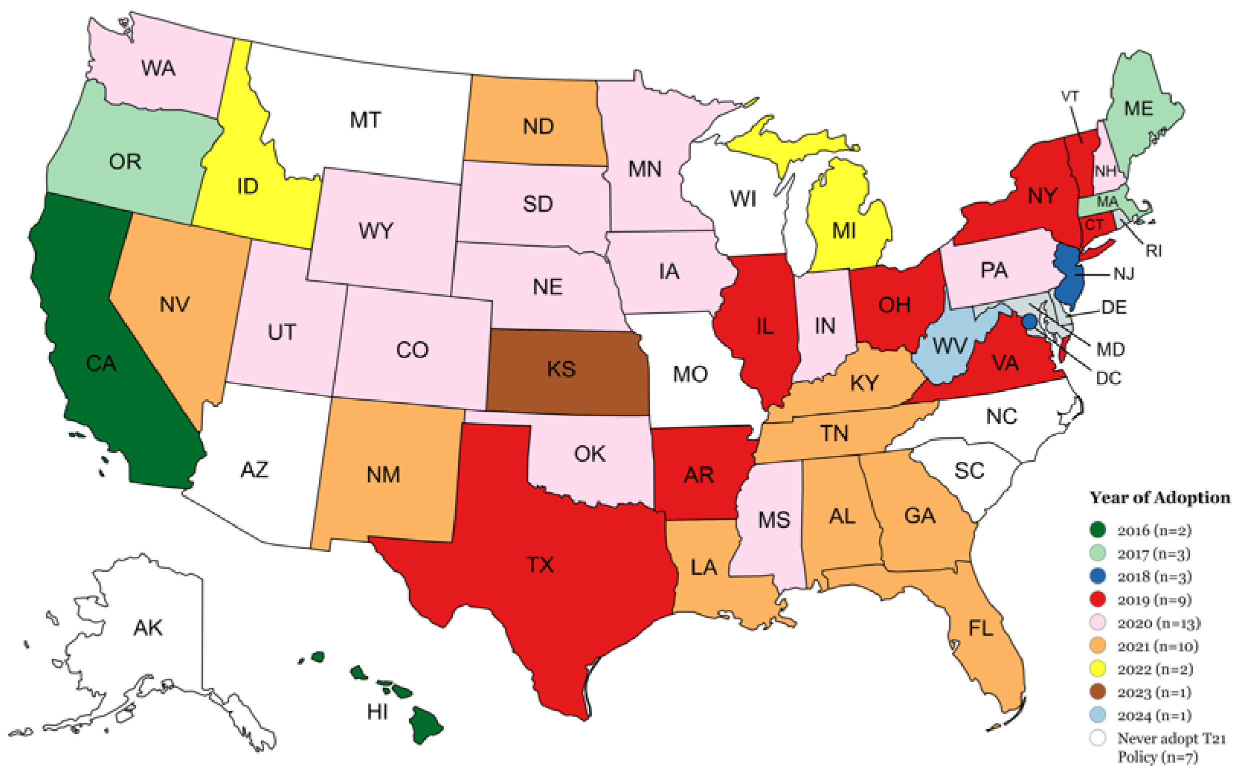
Note: The first-two columns presents the simple average of the event-time effects obtained using the Two-way Fixed effect estimator. While Column 3-6 presents the average treatment effect for the Callaway and Sant'Anna (2021) estimator for not yet treated and never treated controls. Column 3 and 4 display the estimates for the never treated controls, while column 5 and 6 presents that of the not yet treated controls. Young adults are observed at ages 18-20 years between 2011 and 2019 for current Smoking outcome. Unless otherwise stated all regressions control for state, and year fixed effects. Additionally, we control for individual characteristics and other state tobacco policies. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table 4: Heterogeneity Test for Asthma Outcome (Source: BRFSS, 2011 to 2019)

	T21 Policy (1)	Sample Size (2)	Pre-2016 Mean (3)
Full Sample			
	0.05	10,220	0.11
	(0.05)		
By Sex			
Male	0.04	5,555	0.09
	(0.05)		
Female	0.05	4,653	0.13
	(0.10)		
By Race			
White	0.02	6,811	0.11
	(0.06)		
Black	-0.10	1,007	0.13
	(0.23)		
Hispanic	0.17**	1,201	0.09
	(0.08)		
Other race	0.01	734	0.12
	(0.07)		
By Educational Status			
Less than High School	0.23*	1,533	0.13
	(0.14)		
High School	0.08	5,062	0.11
	(0.05)		
Some College	0.01	3,481	0.11
	(0.06)		
By Employment Status			
Working but not Self-employed	0.01	3,687	0.10
	(0.06)		
Self-employed	0.03	336	0.15
	(0.20)		
Not Working	0.12*	6,105	0.12
	(0.06)		
By Age			
18	0.12	4,044	0.11
	(0.10)		
19	-0.06	3,160	0.12
	(0.06)		
20	0.15*	3,010	0.11
	(0.08)		

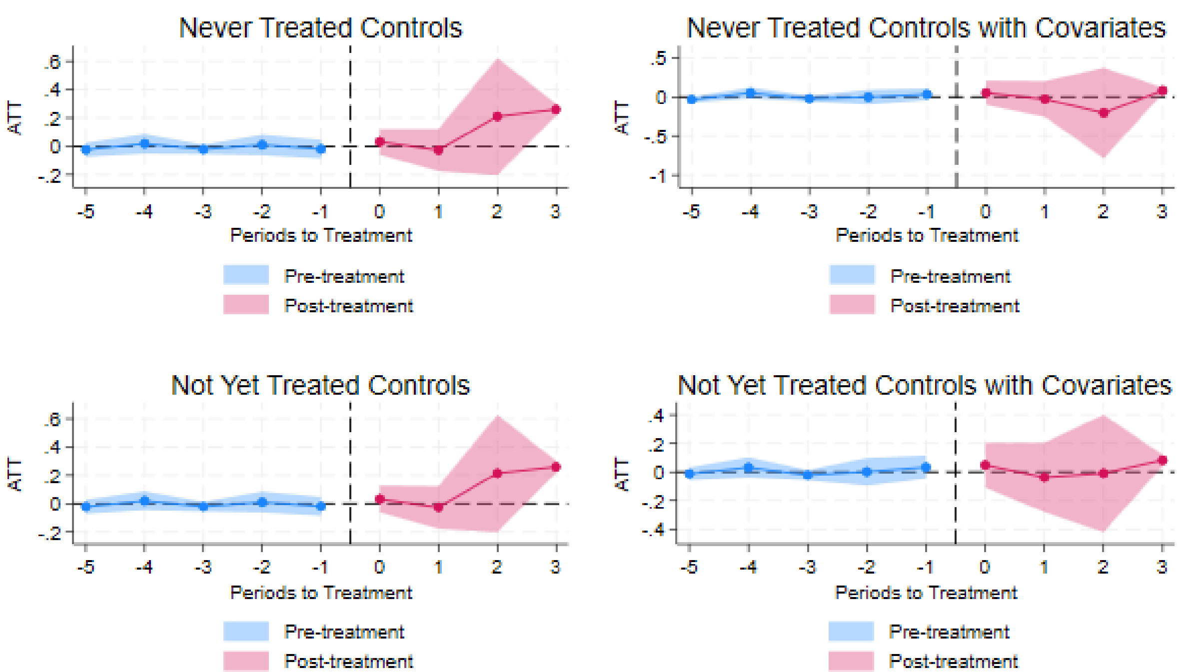
Note: The table presents the heterogeneous test estimates for the individual demographic characteristics using the Callaway and Sant'Anna (2021) DiD estimator with never treated controls. Young adults are observed at ages 18-20 years between 2011 and 2019 for asthma outcome. Unless otherwise stated, all regressions control for state and year-fixed effects. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Figure 1: Timing of Adoption of State Tobacco 21 MLSA Policy



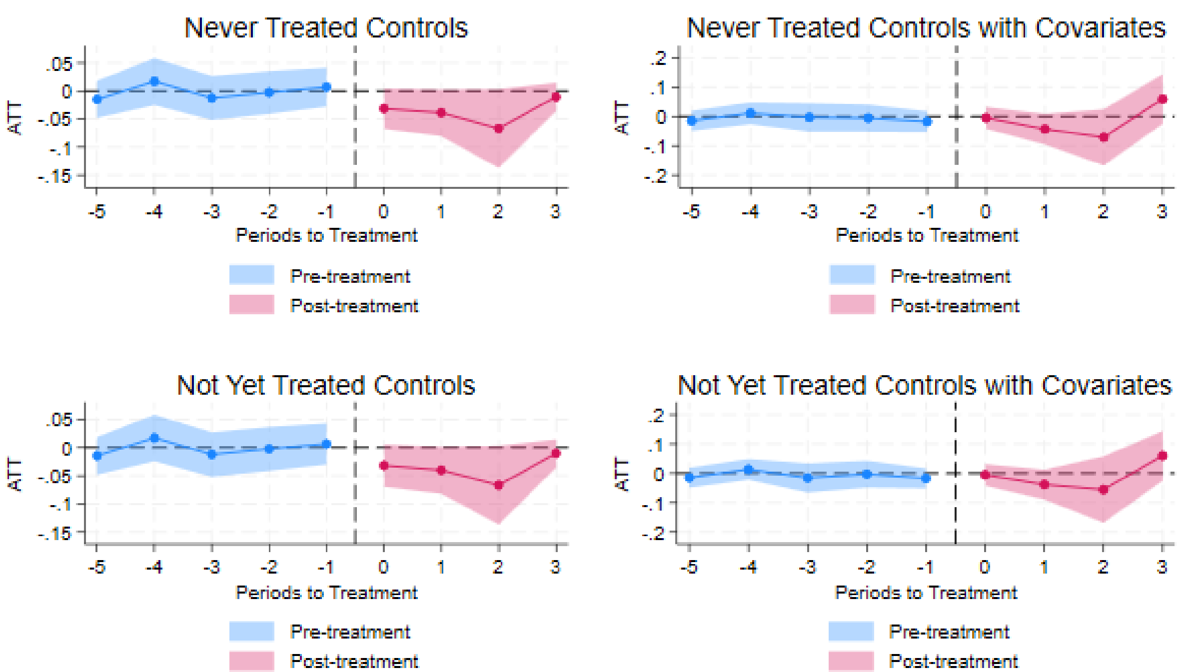
Note: Shows the timing of the adoption of the state-level T21 MLSA policy as of December 2024. However, in the main analysis, we excluded states with local Tobacco 21 policies in place prior to the statewide implementation. Additionally, since our study covers the period between 2011 and 2019, we classified states that adopted the Tobacco 21 MLSA policy after 2019 as control states, along with those that have not yet adopted the policy as of December 2024. We considered a state. to have an effective Tobacco 21 MLSA policy in a given calendar year if it adopted the policy by December of that year. For example, Massachusetts's effective year is coded as 2018, even though the policy took effect on December 31, 2018. See Appendix Table A.1 for the exact month and year of adoption

Figure 2: Callaway and Sant'Anna (2021) Event Study Estimates for Asthma



*Note:*Presents the event study estimates showing the effect of Tobacco 21 MLSA policy on the prevalence of asthma among teenagers and young adults from aged 18 to 20 using the Callaway and Sant'Anna (2021) difference-in-difference estimator. We control for state and year fixed effects for all estimates. Additionally, we control for individual characteristics and other state tobacco policies.

Figure 3: Callaway and Sant'Anna (2021) Event Study Estimates for Current Smokers



*Note:*Presents the event study estimates showing the effect of Tobacco 21 MLSA policy on current smokers among teenagers and young adults from aged 18 to 20 using the Callaway and Sant'Anna (2021) difference-in-difference estimator. We control for state and year fixed effects for all estimates. Additionally, we control for individual characteristics and other state tobacco policies.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT and Grammarly in order to improve the writing of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Appendix

A Tobacco 21 State Laws MLSA

Table A.1: T21 State Laws and their Effective dates as of December, 2024

States	**Effective Date of Tobacco 21 MLSA**
Hawaii	1/1/2016
California	6/9/2016
Washington D.C.	2/18/2017
New Jersey	11/1/2017
Oregon	1/1/2018
Maine	7/1/2018
Massachusetts	12/31/2018
Illinois, Virginia	7/1/2019
Delaware	7/16/2019
Arkansas, Vermont, Texas	9/1/2019
Maryland, Connecticut	10/1/2019
New York	11/13/2019
Ohio	10/16/2019
Washington	1/1/2020
Oklahoma	5/19/2020
Iowa	6/29/2020
Utah, Pennsylvania, South Dakota, Wyoming, Indiana	7/1/2020
Mississippi	7/8/2020
Colorado	7/14/2020
New Hampshire	7/29/2020
Minnesota	8/17/2020
Nebraska	10/1/2020
Kentucky, New Mexico, Tennessee, Georgia	1/1/2021
Nevada	5/27/2021
Rhode Island, Louisiana, Alabama, North Dakota	7/1/2021
Florida	10/1/2021
Idaho	7/1/2022
Michigan	7/21/2022
Kansas	7/1/2023
West Virginia	6/7/2024

Note: The table displays only states that have implemented the Tobacco 21 MLSA law as of December 2024. However, in the main analysis, we excluded states with local Tobacco 21 policies in place prior to the statewide implementation. Additionally, since this study covers the period between 2011 and 2019, we classified states that adopted the Tobacco 21 MLSA policy after 2019 as control states, along with those that have not yet adopted the policy as of December 2024. Source: https://tobacco21.org

B Main Results and Robustness Checks

Table B.1: Effect of T21 MLSA Policy on Asthma for Never Treated Controls

(a) Unconditional Parallel Trends					
	Partially aggregated				Single parameters
TWFE					0.02
					(0.03)
Simple Weighted Average					0.05
					(0.05)
Group Specific effects	g=2016	g=2017	g=2018	g=2019	
	0.10***	0.08	0.06*	-0.00	0.03
	(0.02)	(0.17)	(0.03)	(0.06)	(0.05)
Event study	e=0	e=1	e=2	e=3	
	0.03	-0.03	0.21	0.26***	0.12
	(0.05)	(0.08)	(0.21)	(0.02)	(0.07)
Calendar time effects	t=2016	t=2017	t=2018	t=2019	
	0.22***	-0.06	0.08	0.05	0.07**
	(0.02)	(0.05)	(0.08)	(0.07)	(0.03)
(b) Conditional Parallel Trends					
	Partially aggregated				Single parameters
TWFE					0.02
					(0.03)
Simple Weighted Average					0.01
					(0.08)
Group Specific effects	g=2016	g=2017	g=2018	g=2019	
	-0.32***	0.09	0.15**	0.05	0.04
	(0.03)	(0.13)	(0.06)	(0.11)	(0.07)
Event study	e=0	e=1	e=2	e=3	
	0.05	-0.03	-0.20	0.08***	-0.02
	(0.08)	(0.12)	(0.30)	(0.02)	(0.11)
Calendar time effects	t=2016	t=2017	t=2018	t=2019	
	-0.18***	-0.12	-0.03	0.07	-0.07
	(0.05)	(0.14)	(0.20)	(0.09)	(0.08)

Note: Panel (a) presents the unconditional parallel trends average treatment effect for the two-way fixed effects and Callaway and Sant'Anna (2021) DiD for never-treated controls. Panel (b) presents the conditional parallel trends average treatment effects for the two-way fixed effects and Callaway and Sant'Anna (2021) DiD for never-treated controls where we control for individual characteristics and other state tobacco policies. Young adults are observed at ages 18-20 years between 2011 and 2019 for asthma outcome. Unless otherwise stated, all regressions control for state and year-fixed effects. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table B.2: Effect of T21 MLSA Policy on Asthma for Not Yet Treated Controls

(a) Unconditional Parallel Trends

	Partially aggregated				Single parameters
TWFE					0.02
					(0.03)
Simple Weighted Average					0.05
					(0.05)
Group Specific effects	g=2016	g=2017	g=2018	g=2019	
	0.11***	0.08	0.07**	-0.00	0.03
	(0.01)	(0.17)	(0.03)	(0.06)	(0.05)
Event study	e=0	e=1	e=2	e=3	
	0.03	-0.03	0.21	0.26***	0.12
	(0.05)	(0.08)	(0.21)	(0.02)	(0.07)
Calendar time effects	t=2016	t=2017	t=2018	t=2019	
	0.23***	-0.07	0.09	0.05	0.08**
	(0.02)	(0.05)	(0.08)	(0.07)	(0.08)

(b) Conditional Parallel Trends

	Partially aggregated				Single parameters
TWFE					0.02
					(0.03)
Simple Weighted Average					0.02
					(0.07)
Group Specific effects	g=2016	g=2017	g=2018	g=2019	
	-0.21***	0.08	0.14***	0.05	0.05
	(0.02)	(0.14)	(0.06)	(0.11)	(0.08)
Event study	e=0	e=1	e=2	e=3	
	0.05	-0.04	-0.01	0.08***	0.02
	(0.08)	(0.12)	(0.21)	(0.02)	(0.08)
Calendar time effects	t=2016	t=2017	t=2018	t=2019	
	-0.19***	-0.15	0.08	0.07	-0.05
	(0.03)	(0.14)	(0.12)	(0.09)	(0.06)

Note: Panel (a) presents the unconditional parallel trends average treatment effect for the two-way fixed effects and Callaway and Sant'Anna (2021) DiD for never-treated controls. Panel (b) presents the conditional parallel trends average treatment effects for the two-way fixed effects and Callaway and Sant'Anna (2021) DiD for not yet-treated controls where we control for individual characteristics and other state tobacco policies. Young adults are observed at ages 18-20 years between 2011 and 2019 for asthma outcome. Unless otherwise stated, all regressions control for state and year-fixed effects. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table B.3: Effect of T21 MLSA Policy on Current Smokers for Never Treated Controls

(a) Unconditional Parallel Trends					
	Partially aggregated				Single parameters
TWFE					-0.035**
					(0.013)
Simple Weighted Average					-0.036**
					(0.018)
Group Specific effects	g=2016	g=2017	g=2018	g=2019	
	-0.004	-0.103***	0.007	-0.022	-0.030*
	(0.004)	(0.028)	(0.012)	(0.026)	(0.017)
Event study	e=0	e=1	e=2	e=3	
	-0.031*	-0.039*	-0.067*	-0.010	-0.037**
	(0.018)	(0.021)	(0.036)	(0.013)	(0.017)
Calendar time effects	t=2016	t=2017	t=2018	t=2019	
	-0.009	-0.066**	-0.036	-0.031	-0.035**
	(0.013)	(0.032)	(0.027)	(0.021)	(0.015)
(b) Conditional Parallel Trends					
	Partially aggregated				Single parameters
TWFE					-0.030**
					(0.013)
Simple Weighted Average					-0.018
					(0.020)
Group Specific effects	g=2016	g=2017	g=2018	g=2019	
	0.021	-0.093***	0.007	-0.000	-0.010
	(0.042)	(0.018)	(0.014)	(0.029)	(0.018)
Event study	e=0	e=1	e=2	e=3	
	-0.005	-0.043	-0.071	-0.059	-0.015
	(0.019)	(0.027)	(0.049)	(0.043)	(0.024)
Calendar time effects	t=2016	t=2017	t=2018	t=2019	
	0.058	-0.050***	-0.035	0.010	-0.009
	(0.049)	(0.018)	(0.031)	(0.023)	(0.020)

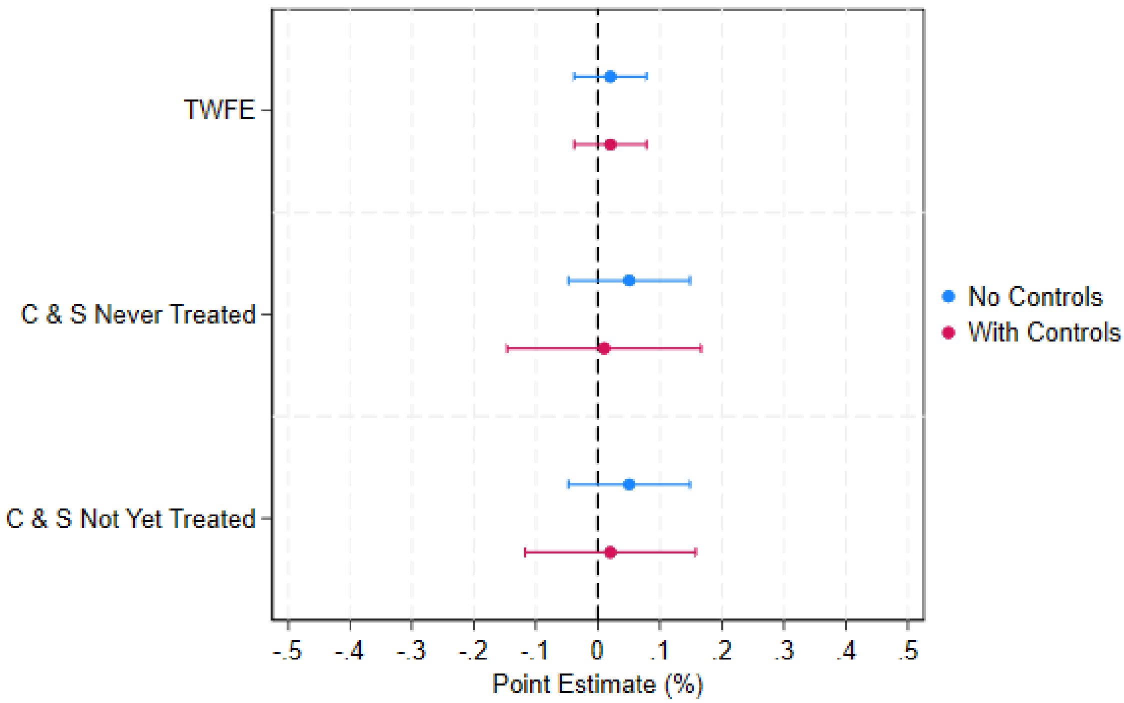
Note: Panel (a) presents the unconditional parallel trends average treatment effect for the two-way fixed effects and Callaway and Sant'Anna (2021) DiD for never-treated controls. Panel (b) presents the conditional parallel trends average treatment effects for the two-way fixed effects and Callaway and Sant'Anna (2021) DiD for never-treated controls where we control for individual characteristics and other state tobacco policies. Young adults are observed at ages 18-20 years between 2011 and 2019 for current smoking outcome. Unless otherwise stated, all regressions control for state and year-fixed effects. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table B.4: Effect of T21 MLSA Policy on Current Smokers for Not Yet Treated Controls

(a) Unconditional Parallel Trends					
		Partially aggregated			Single parameters
TWFE					-0.035**
					(0.013)
Simple Weighted Average					-0.037**
					(0.019)
Group Specific effects	g=2016	g=2017	g=2018	g=2019	
	-0.004	-0.107***	0.005	-0.022	-0.030*
	(0.007)	(0.028)	(0.011)	(0.026)	(0.017)
Event study	e=0	e=1	e=2	e=3	
	-0.032*	-0.040*	-0.067*	-0.010	-0.037**
	(0.019)	(0.021)	(0.036)	(0.013)	(0.017)
Calendar time effects	t=2016	t=2017	t=2018	t=2019	
	-0.007	-0.072**	-0.034	-0.031	-0.036**
	(0.011)	(0.032)	(0.026)	(0.021)	(0.015)
(b) Conditional Parallel Trends					
		Partially aggregated			Single parameters
TWFE					-0.030**
					(0.013)
Simple Weighted Average					-0.016
					(0.021)
Group Specific effects	g=2016	g=2017	g=2018	g=2019	
	0.033	-0.094***	0.009	-0.000	-0.009
	(0.036)	(0.018)	(0.013)	(0.029)	(0.018)
Event study	e=0	e=1	e=2	e=3	
	-0.006	-0.038	-0.056	0.059	-0.010
	(0.019)	(0.026)	(0.058)	(0.043)	(0.025)
Calendar time effects	t=2016	t=2017	t=2018	t=2019	
	0.046	-0.046***	-0.024	-0.010	-0.008
	(0.040)	(0.016)	(0.033)	(0.023)	(0.019)

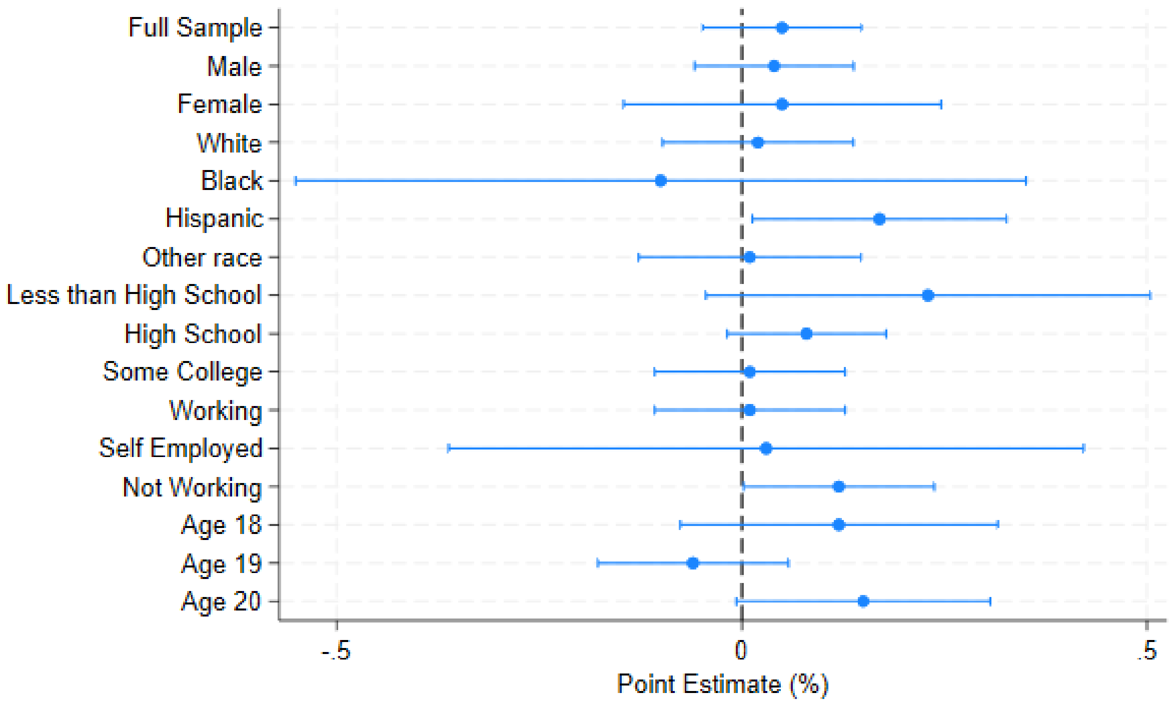
Note: Panel (a) presents the unconditional parallel trends average treatment effect for the two-way fixed effects and Callaway and Sant'Anna (2021) DiD for never-treated controls. Panel (b) presents the conditional parallel trends average treatment effects for the two-way fixed effects and Callaway and Sant'Anna (2021) DiD for never-treated controls where we control for individual characteristics and other state tobacco policies. Young adults are observed at ages 18-20 years between 2011 and 2019 for current smoking outcome. Unless otherwise stated, all regressions control for state and year-fixed effects. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Figure B.1: Main Regression Estimates for Asthma



Note: Callaway and Sant'Anna (2021) Two-Way Fixed Effect Estimates evaluating the impact of Tobacco 21 MLSA policy on the prevalence of asthma among teenagers and young adults aged 18 to 20. The red confidence interval represents estimates with controls, while the blue confidence interval represents estimates without controls.

Figure B.2 : Heterogeneous Test Estimates for Asthma



Note: Presents the heterogeneous test estimates for individuals in the sample using the Callaway and Sant'Anna (2021) difference-in-difference estimator. The regression controls for state and year fixed effects.

Table B.5: Placebo Test for Alternate Outcomes(Source: BRFSS 2011 to 2019)

				Never Treated			Not yet Treated		
	TWFE	TWFE	TWFE	C & S	C & S	C & S	C & S	C & S	C & S
	Health Insurance	Unemployed	HH/Income	Health Insurance	Unemployed	HH/Income	Health Insurance	Unemployed	HH/Income
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
T21Policy	-0.003	0.003	2,443.17	-0.005	0.070	-3,093.92	-0.009	0.072	-3,288.98
	(0.037)	(0.033)	(3,696.29)	(0.045)	(0.055)	(7,838.48)	(0.043)	(0.055)	(7,714.75)
Sample Size	10,066	10,325	10,325	10,066	10,325	10,325	10,066	10,325	10,325
Pre-2016 Mean	0.78	0.53	50,546.22	0.78	0.53	50,546.22	0.78	0.53	50,546.22
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	No	No	No	No	No	No	No	No

Note: Table B.5 presents the placebo test estimates for alternate outcomes using the two-way fixed effects and Callaway and Sant'Anna (2021) DiD estimator with never treated controls. Young adults are observed at ages 18-20 years between 2011 and 2019 for all unrelated outcomes. Unless otherwise stated, all regressions control for state and year-fixed effects. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table B.6 : Falsification Test for Age group 24-28 (Source:BRFSS 2011 to 2019)

			Never Treated		Not yet Treated	
	TWFE	TWFE	C & S	C & S	C & S	C & S
	(1)	(2)	(3)	(4)	(5)	(6)
T21Policy	-0.02	-0.02	0.01	-0.02	0.01	-0.02
	(0.01)	(0.01)	(0.02)	(0.03)	(0.02)	(0.03)
Sample size	26,383	26,383	26,383	26,383	26,383	26,383
Pre-2016 Mean	0.10	0.10	0.10	0.10	0.10	0.10
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes

Note: The first-two columns presents the simple average of the event-time effects obtained using the Two-way Fixed effect estimator. While Column 3-6 presents the average treatment effect for the Callaway and Sant'Anna (2021) estimator for not yet treated and never treated controls. Young adults are observed at ages 24-28 years between 2011 and 2019 for asthma outcome. All regressions exclude states with local T21 MLSA law and control for state, and year fixed effects. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.

Table B.7: Sensitivity Analysis (Source: BRFSS 2011 to 2019)

			Never Treated		Not yet Treated	
	TWFE	TWFE	C & S	C & S	C & S	C & S
	(1)	(2)	(3)	(4)	(5)	(6)
T21Policy	0.05**	0.05**	0.08**	0.14**	0.08**	0.14**
	(0.02)	(0.02)	(0.03)	(0.06)	(0.03)	(0.06)
Sample size	12,338	12,338	12,338	12,338	12,338	12,338
Pre-2016 Mean	0.11	0.11	0.11	0.11	0.11	0.11
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes

Note: The first-two columns presents the simple average of the event-time effects obtained using the Two-way Fixed effect estimator. While Column 3-6 presents the average treatment effect for the Callaway and Sant'Anna (2021) estimator for not yet treated and never treated controls. Young adults are observed at ages 18-20 years between 2011 and 2019 for asthma outcome. The analysis also includes states like CA, IL, MA, MO, NY, and OH which had local Tobacco 21 MLSA laws in place prior to the implementation of statewide policies. Unless otherwise stated all regressions control for state, and year fixed effects. We also control for individual characteristics and other state tobacco policies. ***p<0.01, **p<0.05, *p<0.10. Standard errors are in parentheses.